

SEA4001W: Exercise 8 - Backwards in space

Tristan Mitchell: MTCTRI001

Exercise 8

backward Taylor expansion eq. (20)

$$U(x - \Delta x) = U(x) - \frac{\partial U}{\partial x} \Delta x + \frac{1}{2!} \frac{\partial^2 U}{\partial x^2} \Delta x^2 - \frac{1}{3!} \frac{\partial^3 U}{\partial x^3} \Delta x^3 + \mathcal{O}(\Delta x^4)$$

first order approximation & truncation error

$$U(x - \Delta x) = U(x) - \frac{\partial U}{\partial x} \Delta x + \mathcal{O}(\Delta x^2)$$

$$\frac{\partial U}{\partial x} \Delta x = U(x) - U(x - \Delta x) + \mathcal{O}(\Delta x^2)$$

$$\frac{\partial U}{\partial x} = \frac{U(x) - U(x - \Delta x) + \mathcal{O}(\Delta x^2)}{\Delta x}$$

$$\frac{\partial U}{\partial x} = \frac{U(x) - U(x - \Delta x) + \mathcal{O}(\Delta x)}{\Delta x}$$

reduces to
first-order
accuracy

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discrete indices

$$\begin{aligned} x &= x_i = x_0 + i \Delta x \\ x - \Delta x &= x_{i-1} \\ U(x_i, t_n) &= U_i^n \\ U(x_{i-1}, t_n) &= U_{i-1}^n \end{aligned}$$

point $\rightarrow$ index i time $\rightarrow$ index n
one step backwards in space means
going from index i to $i-1$
both at the same time t_n

$$\therefore \frac{\partial U}{\partial x} \approx \frac{U_i^n - U_{i-1}^n}{\Delta x} \rightarrow \text{upstream}$$